

Educational Effects on Virtual Reality Based Exhibitions of Elementary School Students' Artworks

Se-eun Lim

Department of IT Convergence,
University of Ulsan

Jiyeon Sung

Nexongames

Yujin Park

HD Hyundai ENT

Dongsik Jo[†]School of ICT Convergence,
University of Ulsan

Joonwoo Jeon

Hwaseung Cooperation

ABSTRACT

Recently, virtual reality(VR) technologies have been widely applied in education fields by providing immersive experience. In particular, teachers at the remote sites can deliver realistic and interactive three-dimensional contents to a large number of students. In this paper, we present a VR museum system for elementary school students to visualize and interact with their self-created drawings. The proposed system is a desktop-based metaverse platform that enables virtual interaction. Our system was deployed in several elementary schools, and a survey study was conducted with participating students. Qualitative analysis indicated high levels of interest in avatar-based interactions within the virtual exhibition environment. The results further suggest that the proposed system can enhance educational effectiveness.

Index terms: Education, Virtual Reality, Exhibition, Artworks.

1 INTRODUCTION

In elementary school art education, students engage in creative activities to produce artworks; however, due to limitations in physical space, most of these works are discarded usually and immediately after the class [1]. As a result, students have limited opportunities to revisit their own artworks or to communicate with others about them. Although existing web-based online galleries provide exhibition-like experiences through pre-recorded and visually reconstructed exhibition spaces, they have inherent limitations in conveying a sense of spatial presence and supporting real-time interaction [2]. In contrast, immersive virtual reality(VR) systems can enhance children's artworks experiences [3]. However, Further considerations are required when applying this system to children. For example, head-mounted display(HMD)-based high-immersion VR systems may reduce potential risks related to physical safety and cognitive load for children in developmental stages [4, 5].

Accordingly, we designed and implemented *METASEUM* (the term combined "metaverse" and "museum"), a child-friendly, desktop-based metaverse platform that enables psychological embodiment by allowing user to represent themselves through creative artifacts [6].

[†] e-mail: dongsikjo@ulsan.ac.kr(Correspondence)



Figure 1: (Top) elementary school students' artworks, (Bottom) multi-participant's interaction such as exploring drawings and communicating with each other in VR museum environments.

Figure 1 shows our system configuration, which was deployed in several elementary schools. In the proposed system, students worked collaboratively in groups to create artworks, and the students' drawings were scanned and mapped into texture images, which were subsequently presented through a virtual reality (VR) system. In addition, the system enabled students to view artworks created by other groups and to participate in discussion in shared environments. Then, participants were provided with customizable avatar templates, allowing them to participate in the desktop-based VR environment.

2 STUDY DESIGN AND PROCEDURE

Participants were 98 elementary school students from four schools, consisting of one class each from grades 2 and 3, and two classes from grade 6, corresponding to an age range of 9 to 12 years. The

study was conducted during regular class sessions. First, students received a brief introduction to the metaverse and the METASEUM platform for about 15 minutes, followed by a virtual exhibition experience using desktop or laptop computers for about 20 minutes. Also, students located their own drawings in the virtual space and shared their impressions through text-based chat with friends. After the session, a short questionnaire focusing on interest, ease of use, and classroom applicability was administered for approximately 5 minutes. The questionnaire was designed for elementary school students, using simple wording, and short responses were obtained from teachers for research purposes. In our experiment, ethical review and approval were not required for the study due to minimal risk activity (e.g., drawing and computer-based class without requiring head-mounted displays (HMDs))

3 EXPERIMENT RESULTS

In this study, students conducted a post-experience questionnaire focusing on interest, ease of use, and classroom applicability. Figure 2 is a summary of the survey results. Over 70% (≥ 67) of the participants reported high level of interest and classroom applicability. On the other hand, ease of use was evaluated relatively low, meaning that the VR system requires a more intuitive user interface and improved usability. One participant suggested enabling immediate transmission of photos captured via a smartphone to the desktop-VR system with a single button press. This functionality is identified as a direction for future work. Additional responses indicated that participants recognized their avatars as self-representative autonomous agents and clearly experienced their role as active agents in controlling the virtual environment.

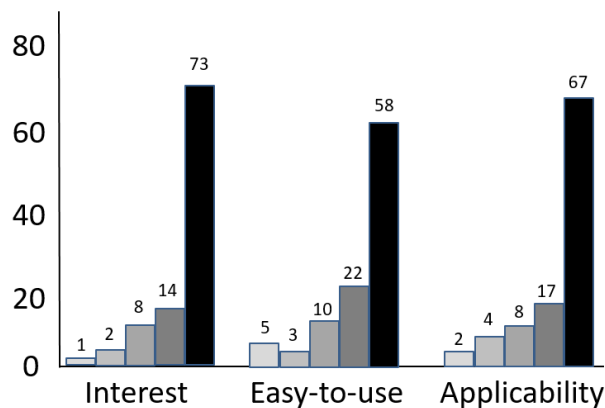


Figure 2: Effectiveness-Based Survey Results, (a) Interest, (b) ease of use, (c) classroom applicability

Overall, the results of the study show that students were actively involved in sharing their work in real time through avatars and communicating with colleagues within virtual exhibition environments, suggesting meaningful participation beyond physical classroom constraints. These interaction patterns highlight the potential role of shared virtual spaces in supporting social presence and agency-related experiences. From the perspective of previous studies on social presence, joint presence can be an important factor in these virtual exhibition environments.

4 CONCLUSIONS AND FUTURE WORKS

This study explored a desktop-based virtual exhibition platform that allows elementary school students to interact with their self-created artworks through avatars in shared virtual environments. The system was designed to provide accessible and psychologically safe virtual experiences for children without relying on immersive VR display devices. The results indicate that students actively engaged in the virtual exhibition by locating, sharing, and discussing their artworks through avatar. These findings provide meaningful evidence that virtual exhibition environments can overcome the constraints of physical classrooms. In our study, we had several limitations. The study was conducted within a single class session, and embodiment-related experiences and long-term learning outcomes were not directly measured. In addition, usability ratings were relatively lower than interest ratings, highlighting the need for more intuitive interaction designs. Future works will investigate children's virtual exhibition experiences through extended usage scenarios, and additional approaches for quantitatively measuring social experience indicators such as co-presence. Also, we will investigate effects of embodiment elements in our constructed VR environment.

ACKNOWLEDGMENTS

This work was supported by the Technology Innovation Program (or Industrial Strategic Technology Development Program - Shipbuilding and Offshore Industry Core Technology Development Program) (RS-2024-00508239, Development of a welding simulator for shipyard worker training based on eXtended reality) funded by the Ministry of Trade, Industry & Energy (MOTIE, Korea) & This research was supported by the MSIT(Ministry of Science and ICT), Korea, under the National Program for Excellence in SW(2024-0-00025) supervised by the IITP(Institute of Information & communications Technology Planning & Evaluation) in 2026.

REFERENCES

- [1] Park S. (2024). A Qualitative Study on the Management and Handling of Student Artworks and Materials in Art classes. *Art Education Research Review*, 38(3), pp.177–208.
- [2] Klingenberg, S., Jørgensen, M. L., Dandanell, G., Skriver, K., Mottelson, A., & Makransky, G. (2020). Investigating the effect of teaching as a generative learning strategy when learning through desktop and immersive VR: A media and methods experiment. *British Journal of Educational Technology*, 51(6), pp.2115-2138.
- [3] Kiltani, K., Groten, R., & Slater, M. (2012). The sense of embodiment in virtual reality. *Presence: Teleoperators and Virtual Environments*, 21(4), pp.373-387.
- [4] Argelaguet, F., Hoyet, L., Trico, M., & Lécuyer, A. (2016). The role of interaction in virtual embodiment: Effects of the virtual hand representation. In *proceedings of 2016 IEEE virtual reality (VR)*. pp.3-10.
- [5] Bailey, J. O., & Bailenson, J. N. (2017). Immersive virtual reality and the developing child. In *Cognitive development in digital contexts*, pp. 181-200.
- [6] Hui, J., Zhou, Y., Oubibi, M., Di, W., Zhang, L., & Zhang, S. (2022). Research on art teaching practice supported by Virtual Reality (VR) technology in the primary schools. *Sustainability*, 14(3), 1246.